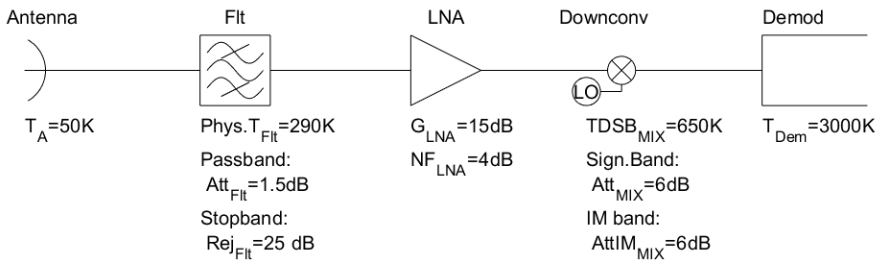
| POLITECNICO MILANO 1863<br>Academic year 2025/2026 | Course<br>RF SYSTEMS | **Student ID or Persona code** |
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| | **Signature** | **Student Name, Surname** |
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# 16 January 2026 Full Exam (26 pts available; ≥15 needed)

Please answer the following questions/problems, providing a brief explanation in English language of the steps/computations involved. Please specify units for all numeric results requiring them. Allowed support material: textbooks, notes, slides, scientific calculator, laptop/tablet (offline).

## Exercise 1 (13 points)

The Earth station in the figure below must receive a signal from a solar probe with bandwidth 30kHz. The relative motion of the probe with respect to Earth however creates a Doppler shift, causing the central frequency to move down at most by 4MHz from the nominal carrier frequency of 8.425GHz. The IF frequency is 330MHz.

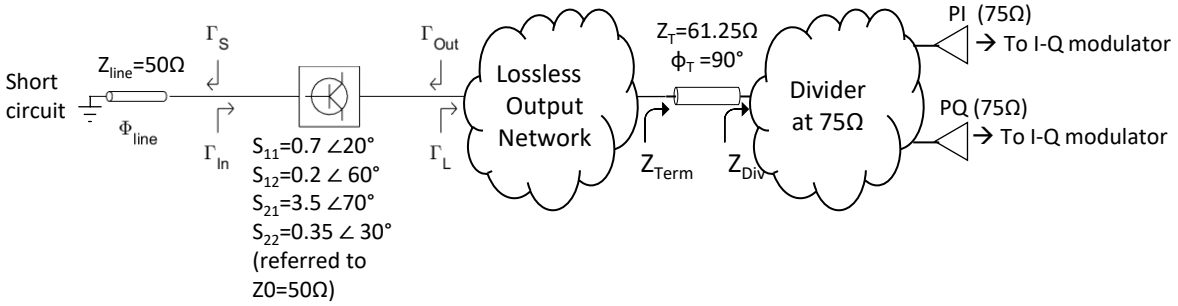



1. In this receiver architecture, what is the purpose of the first filter? Compute the local oscillator frequency range it must cover (below the RF frequency) and the image band edges. Sketch the mask for the fixed bandpass filter and propose reasonable frequencies for its passband and stopband edges.
2. Draw the noise diagram and compute the system noise temperature (at the antenna port) and the corresponding system noise power at the antenna port in the considered 30kHz bandwidth.
3. The transmitter is at a distance of 0.9 million kilometers and transmits an EIRP of 16dBW. To guarantee Eb/N0=5dB, for a data rate of 25kbit/s, what is the required gain of the receiving antenna? (consider the nominal carrier frequency)
4. If the received signal whose average power computed in Question 3 is constituted by two equal tones spaced 20kHz, what is the spectrum at the output of the LNA if it has P1dB=-96.7dBm and a pure third-order model (only within 8.425GHz±100kHz), neglecting noise and compression of the main tones?

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# Exercise 2 (13 points)

An oscillator must be designed to generate a sinusoidal signal at frequency 28GHz. Its output must be then split in two to be the LO signals of an I-Q modulator. **Beware**: the divider and the I-Q modulator are at 75Ohm.




1. Choose and describe a divider (no actual design needed) which can split a signal in two identical versions, with a 90° phase difference between them. In the drawing, show how ports are connected to the rest of the circuit. Remember to add terminations and/or additional delay lines, if needed by your choice.
2. What is the purpose of the $Z_T$ line? Compute the impedance $Z_{Term}$ observed as external termination of the oscillator at 28GHz when both ports PI and PQ are terminated on 75Ohm impedances and the divider is perfectly matched at 75Ohm.
3. Choose/compute $\Gamma_S$ and $\Gamma_L$, imposing $|\Gamma_{out}|$=1.2 and such that the input lossless line has an electric length $\phi_{line}$ between 0° and 90°. Compute the required length $\phi_{line}$ and verify the startup conditions (even if barely satisfied). Use the rule of thumb $z_L$=-re($z_{Out}$)/3-j·Im($z_{Out}$).
4. Assuming a TEM line implements the input $\Gamma_S$ with electrical length $\phi_{line}$ at 28GHz as computed in Question 4, can the oscillation start at 27GHz? (tip: compute the electrical length of the TEM line at this new frequency and check $\Gamma_{out}$)

# Solution Exercise 1

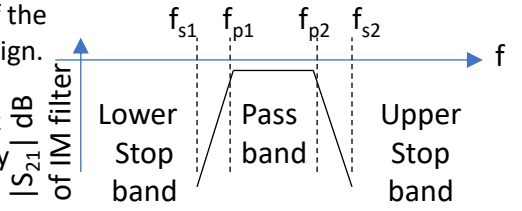
1) The architecture is a standard single-IF receiver (heterodyne) and the first filter acts as image-band filter, to reject primarily potential signals in the image band created by the IF downconversion and to suppress other blockers or interferers which may otherwise saturate the LNA.

The carrier frequency, due to Doppler effect can move around 8.425GHz by [-4;+0]MHz, thus the lowest carrier frequency is fRFmin=8.421GHz while the highest is fRFmax=8.425GHz.

To downconvert these carriers to 330MHz, the Local Oscillator must in fact be between 8.091GHz (=fRFmin-fIF) and 8.095GHz (=fRFmax-fIF).

The image band is located hence at fLO-fIF. Therefore its center can span from 7.761GHz to 7.765GHz.

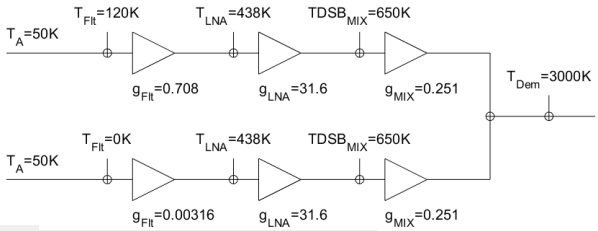
Around the carrier, the signal has a bandwidth of 30kHz. Therefore, the lowest frequency at which we may expect to find useful components is 8.421GHz-30kHz/2=8.420985GHz and the highest frequency is instead 8.425GHz+30kHz/2=8.425015GHz. These is the mandatory passband of the IM filter, although in practice its passband can be larger to ease the design.




Around the image band centers, including the 30kHz bandwidth, the image band edge frequencies are 7.761GHz-30kHz/2=7.760985GHz and 7.765+30kHz/2=7.765015GHz. This image band must be necessarily rejected by the IM filter, hence be in the lower stopband. However, in practice, the stopband can be much larger than this, to ease the design.

Reasonable frequencies for the passband IM filter are thus: fp1=8.3GHz, fp2=8.5GHz, fs1=7.9GHz, fs2=8.9GHz (although this not specified, the closer, the better).

2) The noise diagram is shown on the right, after converting all gains/attenuation to gains in linear scale and NF to Teq=290K($10^{NFdB/10}$-1) and resistive attenuators to Teq=Tphys($10^{AttdB/10}$-1).



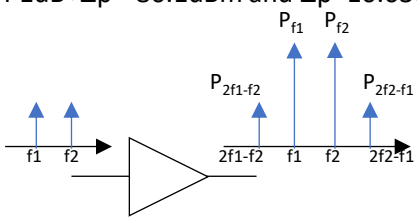

Moving all noise temperatures to the input gives Tsys=2000K:

Tsys= $\left(\frac{T_A}{50K}\right)+\left(\frac{T_{Flt}}{120K}\right)+\left(\frac{T_{LNA}}{438K}\right)/\left(\left(\frac{g_{Flt}}{0.708}\right)\right)+\left(\frac{TDSB_{MIX}}{650K}\right)/\left(\left(\frac{g_{Flt}}{0.708}\right)\cdot\left(\frac{g_{LNA}}{31.6}\right)\right)+\left(\frac{T_{Dem}}{3000K}\right)/\left(\left(\frac{g_{Flt}}{0.708}\right)\cdot\left(\frac{g_{LNA}}{31.6}\right)\cdot\left(\frac{g_{MIX}}{0.251}\right)\right)$
$+\left(\left(\left(\left(\frac{T_A}{50K}\right)\right)\cdot\left(\frac{g_{Flt}}{0.00316}\right)+\left(\frac{T_{LNA}}{438K}\right)\right)\cdot\left(\frac{g_{LNA}}{31.6}\right)+\left(\frac{TDSB_{MIX}}{650K}\right)\right)\cdot\left(\frac{g_{MIX}}{0.251}\right)/\left(\frac{g_{MIX}}{0.251}\right)/\left(\left(\frac{g_{Flt}}{0.708}\right)\cdot\left(\frac{g_{LNA}}{31.6}\right)\right)+\left(\frac{T_{Dem}}{3000K}\right)/\left(\left(\frac{g_{Flt}}{0.708}\right)\cdot\left(\frac{g_{LNA}}{31.6}\right)\cdot\left(\frac{g_{MIX}}{0.251}\right)\right)$

Hence to Pnsys=kTsysB=0.828fW=-120.8dBm at the ant. port.

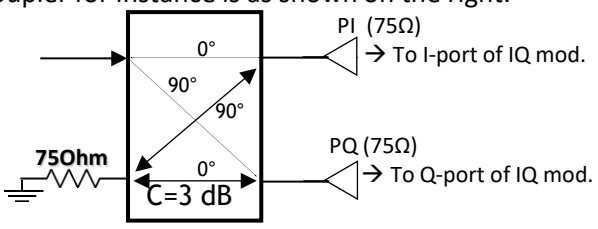
3) Eb/N0=5dB implies that the required SNR is SNR=(Eb/N0) · (R/B)=2.64=4.2dB. Therefore, the useful signal at the antenna port must be at least PRx=-120.8dBm+4.2dB=-116.6dBm. Then, we use Friis eq. for the link budget: PRx=EIRPdBm-FSL+GRx with FSL=20Log(4 π d/λ)=230.0dB → GRx=PRx-EIRPdBm+FSL=67.4Bi

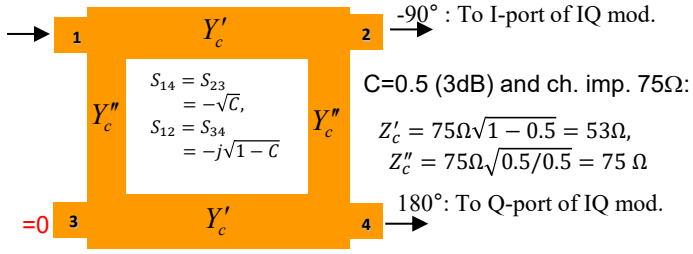
4) The signal has average power PRx=-116.6dBm at the antenna port, hence the tones at ±10kHz from 8.425GHz each are -119.6dBm. After attenuation, the remaining -121.1dBm tones are amplified, thus providing Pf1=Pf2=-106.1dBm at its output. The 3rd order non-linear effect creates intermodulation lines at 2f2-f1=8.425GHz+10kHz+20kHz and 2f1-f2=8.425GHz-10kHz-20kHz with average power each $P_{2f2-f1}=P_{2f1-f2}=3\cdot P_{\omega1}-2\cdot IP3$=-146.1dBm (Carrier-to-intermodulation ratio=40dB), where IP3=P1dB+Δp=-86.1dBm and Δp=10.63dB.



# Solution Exercise 2

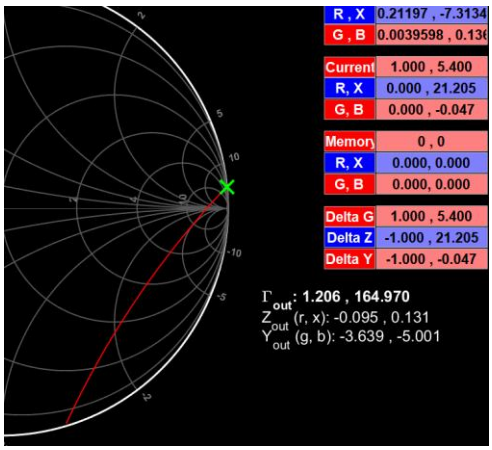
1) Several networks can be devised to split a signal in two equal-amplitude versions with phase difference 90°
Hybrid 3dB directional couplers can indeed be used, as they natively provide a phase difference of 90°. A branch-line coupler for instance is as shown on the right.




2) Whatever choice of divider in Question 1, the divider was required to be designed matched at 75Ohm, and therefore, by definition, when its ports are terminated on 75Ohm, it provides an input impedance of $Z_{div}$=75Ohm (in fact, its scattering parameters referred to its design impedance 75Ohm include S11=0). The $Z_T$, $\Phi_T$ line hence sees 75Ohm as its termination. Since it has an electrical length of 90°, it behaves as imp. Inverter $Z_{term}=Z_T^2/Z_{div}$=50Ohm. (In fact, $\Gamma_{term@ZT}=\Gamma_{div@ZT}e^{-2j\phi T}=-\Gamma_{div@ZT}\rightarrow Z_{div}=Z_T(1+\Gamma_{term@ZT})/(1-\Gamma_{term@ZT})$ $\rightarrow Z_{div}=Z_T(1-\Gamma_{div@ZT})/(1+\Gamma_{div@ZT})$ with $\Gamma_{div@ZT}=(Z_{div}-Z_T)/(Z_{div}+Z_T)$ $\rightarrow Z_{div}= Z_T(2Z_T)/(2Z_{div})=Z_T^2/Z_{div}$.)

3) Drawing the "Map Γ Source" circle with |Γout|=1.2 and knowing that Γs can only be in the top half of the unit circle, only one value is acceptable for Γs=1 ∠ 5.3°. The corresponding output norm. impedance is obtained: zOut=-0.096+j0.131. Then we select zL=0.032-j0.131 and compute the corresponding |Γin|=1.033. Since also |Γin Γs|>1 the oscillation can start (although barely; in practice more margin would be needed).
To achieve the required Γs=1 ∠ 5.3°, the input line must provide rotation from the short circuit (Γshortcircuit=-1) up to 5.3°, which implies a clockwise rotation of 180°-5.3°=174.7° requiring a line 87.35° long.




4) The elec. length of the input line is ϕline=βLline=ω Lline/v
In a TEM line v does not depend on frequency.
At 28GHz we had ϕline=87.35°=1.524rad=2 π 28GHz Lline/v.
At 27GHz: $\phi_{line27}$= 2 π 27GHz Lline/v=27 · 87.35°/28°=84.2°.
Therefore $\Gamma_{s27}=\Gamma_{shortcircuit}e^{-2j\phi line27}$=1 ∠ -335.6°=1 ∠ 11.6°.
This value is barely outside of the stability region for the source (a graphic artefact may look like it is inside), and in fact the corresponding |Γout| <1.
In fact, Γout=0.981 ∠ 175°, therefore the corresponding output impedance has a real part re(zout)=0.0096>0,
which is not acceptable for a negative-resistance oscillator.

